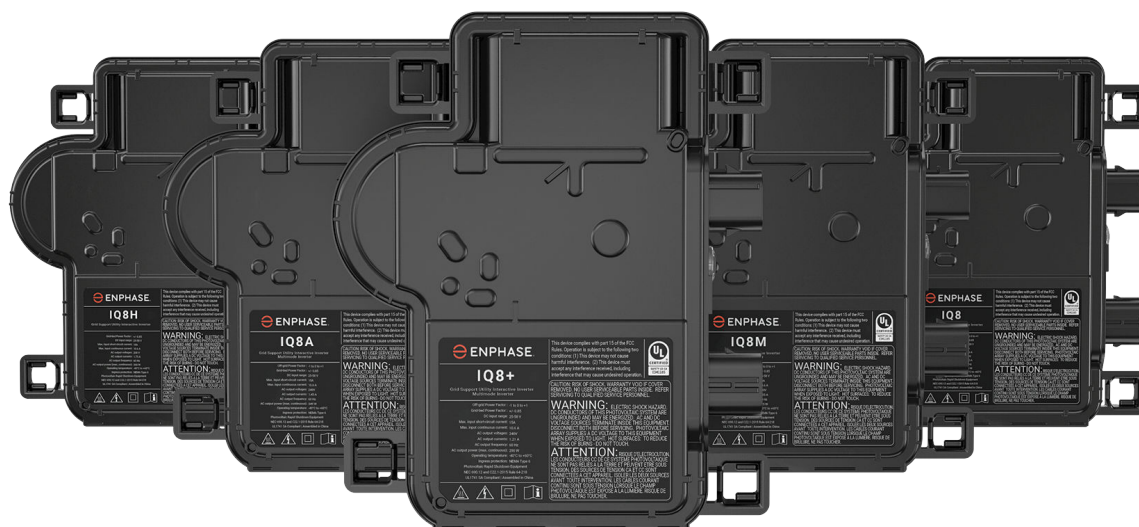


IQ8, IQ8+, IQ8M, IQ8A, and IQ8H Microinverters



Applicable regions: North America

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FCC compliance

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, you are encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and the receiver.
- Connect the equipment to an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Changes or modifications not expressly approved by the party responsible for compliance may void the user's authority to operate the equipment.

Other information

Product information is subject to change without notice. All trademarks are recognized as the property of their respective owners.

User documentation is updated frequently; check the Enphase website for the latest information.
<https://enphase.com/installers/resources/documentation>.

To ensure optimal reliability and to meet warranty requirements, the Enphase microinverter must be installed according to the instructions in this manual. For warranty text, refer to enphase.com/installers/resources/warranty.

For Enphase patent information, refer to <https://enphase.com/patents>.

Audience

This manual is intended for use by professional installation and maintenance personnel.

Important safety information

Read this first

This manual contains important instructions for use during the installation and maintenance of the IQ8 Series Microinverters.



IMPORTANT: Enphase IQ Series Microinverters require the IQ Cable. An IQ Gateway is required to monitor the performance of the IQ Microinverters and configure them to be compliant with certain Authority Having Jurisdictions (AHJs). The IQ accessories work only with Enphase IQ Series Microinverters.

Product labels

The following symbols appear on the product label and are described here.



Hot surface



Refer to safety instructions



Risk of electric shock



Refer to the manual



Double insulated

Safety and advisory symbols

To reduce the risk of electric shock and to ensure the safe installation and operation of the IQ8 Series Microinverters System, the following safety symbols appear throughout this document to indicate dangerous conditions and important safety instructions.



DANGER: This indicates a hazardous situation, which, if not avoided, will result in death or serious injury.



WARNING: This indicates a situation where failure to follow instructions may be a safety hazard or cause equipment malfunction. Use extreme caution and follow instructions carefully.



WARNING: This indicates a situation where failure to follow instructions may result in burn injury



NOTE: This indicates information that is very important for optimal system operation. Follow instructions closely.

IQ8 Series Microinverters safety instruction

General safety



DANGER: Risk of electric shock. Risk of fire.

Only use electrical system components approved for wet locations.

Only qualified personnel should install, troubleshoot, or replace Enphase microinverters or IQ Cable and accessories.

Ensure that all AC and DC wiring is correct and that none of the AC or DC cables are pinched, shorted, or damaged. Also, ensure that all AC junction boxes are properly closed.

Do not exceed the maximum number of microinverters in an AC branch circuit listed in the manual. You must protect each microinverter AC branch circuit with a 20 A maximum breaker or fuse, as appropriate.

Do not use Enphase equipment in a manner not specified by the manufacturer. Doing so may cause death or injury to persons or damage to equipment.

Be aware that installation of this equipment includes the risk of electric shock.

The DC conductors of this photovoltaic system are ungrounded and may be energized.

Always de-energize the AC branch circuit before servicing. Never disconnect the DC or AC connectors under load.



WARNING: Before installing or using the Enphase microinverter, read all instructions and cautionary markings in the technical description of the Enphase equipment and the photovoltaic (PV) equipment.

Do not connect Enphase microinverters to the grid or energize the AC circuit(s) until you have completed all the installation procedures and have received approval from the electrical utility.

When the PV array is exposed to light, DC voltage is supplied to the power conversion equipment (PCE).

Risk of equipment damage. Enphase male and female connectors must only be mated with the identical type and brand of male/female connector.



NOTE: To ensure optimal reliability and to meet warranty requirements, install the Enphase equipment according to the instructions in this manual. Protection against lightning and resulting voltage surges must be in accordance with local standards. Perform all electrical installations in accordance with all applicable local electrical codes, such as the Canadian Electrical Code, Part 1; ANSI requirements; and NFPA 70 (NEC).



DANGER: Risk of electric shock. Always de-energize the AC branch circuit before servicing. Never disconnect the DC or AC connectors under load. When the system has an IQ System Controller, then the RSD must be switched to the OFF/open position to de-energize the AC PV circuit.

Microinverter safety



WARNING: Risk of skin burn.

The chassis of the Enphase microinverter is the heat sink. Under normal operating conditions, the temperature could be 20°C above ambient temperature, but under extreme conditions, the microinverter can reach a temperature of 90°C. To reduce the risk of burns, use caution when working with microinverters.



DANGER: Risk of fire.

The DC conductors of the PV module must be labeled PV Wire or PV Cable when paired with the Enphase microinverter.



DANGER: Risk of electric shock. Risk of fire.

Only qualified personnel may connect the Enphase microinverter to the utility grid.

Do not attempt to repair the Enphase microinverter; it contains no user-serviceable parts. If it fails, contact Enphase Support at <https://enphase.com/contact/support> to obtain a return merchandise authorization (RMA) number and start the replacement process. Tampering with or opening the Enphase microinverter will void the warranty.



WARNING: Risk of equipment damage.

Install the microinverter under the PV module to avoid direct exposure to rain, UV, and other harmful weather events. Always install the microinverter bracket side up. Do not mount the microinverter upside down. Do not expose the AC or DC connectors (on the IQ Cable, PV module, or microinverter) to rain or condensation before the connectors are mated.

The maximum open circuit voltage of the PV module must not exceed the specified maximum input DC voltage of the Enphase microinverter. Refer to <https://enphase.com/en-us/support/module-compatibility> to verify PV module electrical compatibility with the microinverter. Use IQ8 Series Microinverters only with compatible PV modules per the Enphase compatibility calculator. Using an electrically incompatible PV module voids Enphase warranty.

You must match the DC operating voltage range of the PV module with the allowable input voltage range of the Enphase microinverter.

The Enphase microinverter is not protected from damage due to moisture trapped in cabling systems. Never mate microinverters to cables left disconnected and exposed to wet conditions. This voids the Enphase warranty.

The Enphase microinverter functions only with a standard, compatible PV module with appropriate fill factor, voltage, and current ratings. Unsupported devices include smart PV modules, fuel cells, wind or water turbines, DC generators, non-Enphase batteries, etc. These devices do not behave like standard PV modules, so operation and compliance are not guaranteed. These devices may also damage the Enphase microinverter by exceeding its electrical rating, making the system unsafe.



NOTE: The Enphase microinverter has field-adjustable voltage and frequency trip points that may need to be set, depending upon local requirements. Contact Enphase Support at <https://enphase.com/contact/support> for required modifications as per local requirements.

Microinverter safety



NOTE: Installers must check the manufacturing date of the products to ensure that the installation date is within one year of the manufactured date of the products. Contact your local distributor to validate the date code.

IQ Cable safety



DANGER: Risk of electric shock.

Do not install the IQ Terminator while the power is connected.



WARNING: Risk of electric shock. Risk of fire.

When stripping the sheath from the IQ Cable, make sure the conductors are not damaged. If the exposed wires are damaged, the system may not function properly.

Do not leave AC connectors on the IQ Cable uncovered for an extended period. You must cover any unused connector with an IQ Sealing Cap.

Make sure a protective IQ Sealing Cap has been installed on all unused AC connectors. Unused AC connectors are live when the system is energized.

Use the IQ Terminator only once. If you open the IQ Terminator following installation, the latching mechanism is destroyed. If the latching mechanism is defective, do not use the IQ Terminator. Do not circumvent or manipulate the latching mechanism.

When installing the IQ Cable, secure any loose cable to minimize tripping hazards.



NOTE: When looping the IQ Cable, do not form loops smaller than 4.75" (12 cm) in diameter.

Provide support for the IQ Cable every 1.8 m (6 ft).

If you need to remove the IQ Sealing Cap, use the IQ Disconnect Tool.

When installing the IQ Cable and accessories, adhere to the following:

- Do not expose the IQ Terminator cap or cable connections to directed, pressurized liquid (water jets, and so on).
- Do not expose the IQ Terminator or IQ Cable to continuous immersion.
- Do not expose the IQ Terminator cap or IQ Cable connections to continuous tension (for example, tension due to pulling or bending the cable near the connection).
- Use only the connectors provided.
- Do not allow contamination or debris in the connectors.
- Use the IQ Terminator cap and cable connections only when all parts are present and intact.
- Do not install or use in potentially explosive environments.
- Do not allow the IQ Terminator to come into contact with an open flame.
- Fit the IQ Terminator cap using only the prescribed tools and in the prescribed manner.
- Only the IQ Terminator is used to seal the conductor end of the IQ Cable; no other method is allowed.

PV rapid shutdown equipment (PVRSE)

This product is UL Listed as PV rapid shutdown equipment and conforms with NEC-2014, NEC-2017, and NEC-2020 section 690.12 and C22.1-2018 Rule 64-218 rapid shutdown of PV systems, for AC and DC conductors, when installed according to the following requirements:

- Microinverters and all DC connections must be installed inside the array boundary. Enphase further requires that the microinverters and DC connections be installed under the PV module to avoid direct exposure to rain, UV, and other harmful weather events.
- The array boundary is defined as 305 mm (1 ft.) from the array in all directions or 1 m (3 ft.) from the point of entry inside a building.

This rapid shutdown system must be provided with an initiating device or status indicator installed in a location accessible to first responders. Alternately, it must be connected to an automatic system that initiates rapid shutdown in the event a system disconnects or another emergency system is activated.

The initiator shall be listed and identified as a disconnecting means that plainly indicates whether it is in the “off” or “on” position.

Examples are:

- Service disconnecting means
- PV system disconnecting means
- Readily accessible switch or circuit breaker

The handle position of a switch or circuit breaker is suitable for use as an indicator. Refer to NEC or CSA C22.1-2018 for more information.

Additionally, in a prominent location near the initiator device, a placard or label must be provided with a permanent marking including the following wording:

PHOTOVOLTAIC SYSTEM EQUIPPED WITH RAPID SHUTDOWN". The term **PHOTOVOLTAIC** may be replaced with **PV**.

The placard, label, or directory shall be reflective, with all letters capitalized and having a minimum height of 9.5 mm (3/8") in white on a red background.

1. The Enphase IQ System

The Enphase IQ System includes:

- IQ8, IQ8+, IQ8M, IQ8A, and IQ8H Microinverters: The smart grid-ready IQ Series Microinverters convert the DC output of the PV module into grid-compliant AC power.
- IQ Gateway (ENV-IQ-AM1-240): The IQ Gateway is a communication device that provides network access to the PV array. The IQ Gateway collects production and performance data from the IQ Microinverters over on-site AC power lines and transmits the data to Enphase through an internet or cellular modem connection. IQ Gateway is required to update the grid profile setting, which is required for certain utilities. The IQ Gateway is capable of monitoring up to 600 Enphase IQ Microinverters and up to 39 Enphase IQ Batteries. Microinverters should be continuously connected to the internet using an IQ Gateway. This will help ensure that they can be monitored remotely and that the product can receive over-the-air firmware updates.



NOTE: All 208 VAC installations require that you use the Enphase IQ Gateway to commission the Enphase microinverters to propagate correct grid profile settings for 208 VAC trip points.

- Enphase Installer Platform Web-based monitoring and management software: Installers can use the Enphase Installer Platform to view detailed performance data, manage multiple PV systems, and remotely resolve issues that might impact system performance. Find out more at <https://enphase.com/installers/apps>.
- Enphase Installer App for iOS and Android devices: It allows installers to configure the system while on-site, eliminating the need for a laptop and improving installation efficiency. You can use the app to:
 - Connect to the IQ Gateway over a wireless network for faster system setup and verification.
 - View and email a summary report that confirms a successful installation.
 - Scan device serial numbers and sync system information with the Enphase Installer Platform.
- Enphase IQ Battery
- IQ Field Wireable Connectors (Q-CONN-10F and Q-CONN-10M): Make connections from any IQ Cable or open IQ Field Wireable Connector.

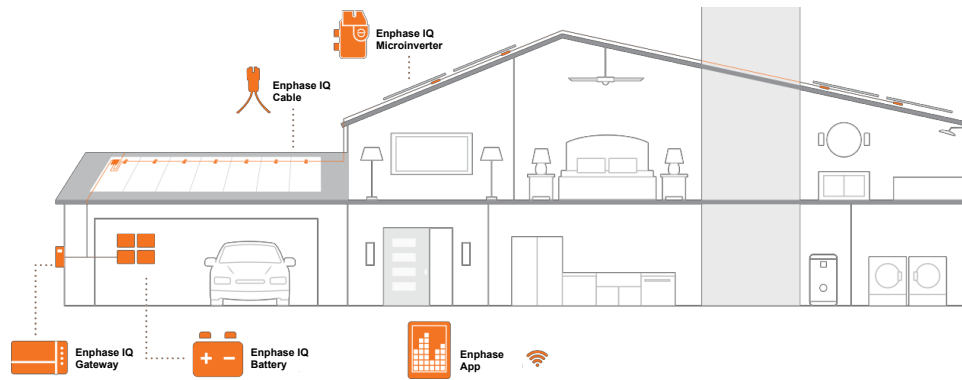
This manual describes the safe installation and operation of the Enphase microinverter.



NOTE: To ensure optimal reliability and to meet warranty requirements, the microinverter must be installed according to the instructions in this manual.

1.1 How the Enphase IQ Series Microinverters work

The Enphase microinverter maximizes energy production by using a sophisticated maximum power point tracking (MPPT) algorithm. Each microinverter individually connects to one PV module in your array. This configuration enables an individual MPPT to control each PV module, ensuring that the maximum power available from each PV module is exported to the utility grid regardless of the performance of the other PV modules in the array. While an individual PV module in the array may be affected by shading, soiling, orientation, or PV module mismatch, each microinverter ensures top performance for its associated PV module.



1.1.1 System monitoring

Once you install the IQ Gateway and provide an internet connection through a broadband router or modem, the IQ Microinverters automatically begin reporting to the Enphase Installer Platform. The Enphase Installer Platform presents current and historical system performance trends and informs you of PV system status.

1.1.2 Optimal reliability

Microinverter systems are inherently more reliable than traditional inverters. The distributed nature of a microinverter system ensures that there is no single point of system failure in the PV system. Enphase microinverters are designed to operate at full power at ambient temperatures as high as 60°C (140°F).

1.1.3 Ease of design

PV systems using Enphase microinverters are very simple to design and install. You will not need string calculations or cumbersome traditional inverters. You can install individual PV modules in any combination of PV module quantity, type, age, and orientation. Each microinverter quickly mounts on the PV racking directly beneath each PV module. Low-voltage DC wires connect from the PV module directly to the co-located microinverter, eliminating the risk of personnel exposure to dangerously high DC voltage.

2. Planning for microinverter installation

The IQ8 Microinverter is compatible with 60-cell PV modules, and the IQ8+, IQ8M, IQ8A, and IQ8H Microinverter support PV modules with 54, 60, 66, or 72-cell PV modules. All of them install quickly and easily. The microinverter housing is designed for outdoor installation and complies with the NEMA 250, type 6 environmental enclosure rating standard.



NOTE: NEMA 6 rating definition: Indoor or outdoor use primarily to provide a degree of protection against hose-directed water, the entry of water during occasional temporary submersion at a limited depth, and damage from external ice formation.

The IQ Cable is available with connector spacing options to accommodate the installation of PV modules in portrait or landscape orientation. For IQ Cable ordering information, see [Enphase IQ Cable planning and ordering](#).

2.1 Compatibility

The IQ Series Microinverters are electrically compatible with PV modules, as listed in the following table. For specifications, see [Technical data](#) on page 33. You can refer to the Enphase Compatibility Calculator at <https://enphase.com/installers/microinverters/calculator> electrical compatibility. To ensure mechanical compatibility, be sure to order the correct connector type for both the microinverter and PV module from your distributor.



WARNING: Risk of fire. The PV module DC conductors must be labeled “PV Wire” or “PV Cable” to comply with NEC for ungrounded PV power systems.

Microinverter model	Connector type	PV module cell count
IQ8-60-2-US	MC4	To meet compatibility, PV modules must be within the following maximum input DC voltage and maximum module I_{sc} . Module compatibility can be checked at https://enphase.com/installers/microinverters/calculator .
IQ8PLUS-72-2-US		
IQ8M-72-2-US		
IQ8A-72-2-US		
IQ8H-240-72-2-US		
IQ8H-208-72-2-US		



NOTE: IQ8 Series Microinverters can be added to existing IQ7 systems on the same IQ Gateway only in the following grid-connected configurations: (i) Solar Only or (ii) Solar + Battery (IQ Battery 3T/10T and IQ Battery 5P) without backup.



NOTE: IQ7 Series Microinverters cannot be added to a site that already has IQ8 Series Microinverters connected to the same IQ Gateway.



NOTE: Adding IQ8 Series Microinverters to an existing IQ7 system with an IQ System Controller is not supported if they are connected to the same IQ Gateway.

2.2 Grounding considerations

The microinverter models listed in this guide do not require grounding electrode conductors (GEC), equipment grounding conductors (EGC), or grounded conductors (neutral). Your Authority Having Jurisdiction (AHJ) may require you to bond the mounting bracket to the racking. If so, use UL2703 hardware or star washers. The microinverter itself has a Class II double-insulated rating, which includes ground fault protection (GFP). To support GFP, use only PV modules equipped with DC cables labeled PV Wire or PV Cable.

2.3 Branch circuit capacity

Plan your AC branch circuits to meet the following limits for a maximum number of microinverters per branch when protected with a 20 A (maximum) overcurrent protection device (OCPD).

Maximum ¹ IQ8 Microinverters per AC branch circuit (240 VAC)	Maximum ¹ IQ8+ Microinverters per AC branch circuit (240 VAC)	Maximum ¹ IQ8M Microinverters per AC branch circuit (240 VAC)
16	13	11
Maximum ¹ IQ8A Microinverters per AC branch circuit (240 VAC)	Maximum ¹ IQ8H Microinverters per AC branch circuit (240 VAC)	Maximum ¹ IQ8H Microinverters per AC branch circuit (208 VAC)
11	10	9

¹Limits may vary. Refer to local requirements to define the number of microinverters per branch in your area.



NOTE: For a mixed system with both IQ7 and IQ8 Microinverters on the same branch, ensure that the total maximum continuous output current of all microinverters on the branch does not exceed 16 A.

2.4 Utility service requirements

The IQ Series Microinverters work with single-phase service. Measure AC line voltages at the electrical utility connection to confirm that it is within the ranges shown in the following table.

240 VAC, single-phase	
L1 to L2	211 to 264 VAC
L1, L2 to ground	106 to 132 VAC
208 VAC, single-phase	
L1 to L2	183 to 229 VAC
L1, L2 to ground	106 to 132 VAC



NOTE: All 208 VAC installations require that you use the Enphase IQ Gateway to commission the Enphase microinverters to propagate correct grid profile settings for 208 VAC trip points.

2.5 Wire lengths and voltage rise

When planning the system, you must select the appropriate AC conductor size to minimize voltage rise. Select the correct wire size based on the distance from the beginning of the microinverter AC branch circuit to the breaker in the load center. Enphase recommends a voltage rise total of less than 2% for the sections from the microinverter AC branch circuit to the breaker in the load center.

Enphase provides guidance about choosing wire size and maximum conductor lengths in the [Voltage rise technical brief](#). Refer to this document for voltage rise values in IQ Cables and on how to calculate voltage rise in other wire sections of the system.

Standard guidelines for voltage rise on feeder and AC branch circuit conductors might not be sufficient for microinverter AC branch circuits that contain the maximum allowable microinverters. This is due to the high inherent voltage rise on the AC branch circuit.



NOTE: The best practice is to center-feed the branch circuit to minimize voltage rise in a fully populated branch. This practice greatly reduces the voltage rise as compared with an end-fed

branch. To center-feed a branch, divide the circuit into two sub-branch circuits protected by a single OCPD.

2.6 Lightning and surge suppression

Enphase microinverters have integral surge protection, greater than most traditional inverters. However, if the surge has sufficient energy, the protection built into the microinverter can be exceeded, and the equipment can be damaged. For this reason, Enphase recommends that you protect your system with a lightning and/or surge suppression device. In addition to having some level of surge suppression, it is also important to have insurance that protects against lightning and electrical surges. Enphase has tested the following devices:

- [Leviton 51110-SRG](#)
- [Schneider SquareD HEPD50](#)



NOTE: Protection against lightning and resulting voltage surges must be in accordance with local standards.

3. Parts and tools required

In addition to the microinverters, PV modules, and racking, you will need the following.

3.1 Enphase equipment

- IQ Gateway: Required to monitor production and configure microinverters to be compliant with certain AHJs. IQ Gateway (model ENV-IQ-AM1-240) communications gateway or IQ Combiner (model X-IQ-AM1-240-B, X-IQ-AM1-240-2, X-IQ-AM1-240-3, X-IQ-AM1-240-3C).
- Enphase Installer App: Download the Enphase Installer App, open it, and log in to your Enphase Installer Platform account. Use it later to scan microinverter serial numbers and connect to the IQ Gateway to track system installation progress. To download, go to <https://enphase.com/installers/apps> or scan the QR code to download it.
- Tie wraps or IQ Cable Clips (Q-CLIP-100)
- IQ Sealing Caps (Q-SEAL-10) for any unused drops on the IQ Cable.
- IQ Terminator (Q-TERM-10) typically two needed per branch circuit.
- IQ Disconnect Tool (Q-DISC-10)
- IQ Field Wireable Connectors (male and female: Q-CONN-10M and Q-CONN-10F)
- IQ Cable:

Cable model	Connector spacing	PV module orientation	Connector count per box
Q-12-10-240	1.3 m	Portrait	240
Q-12-17-240	2.0 m	Landscape (60-cell)	240
Q-12-20-200	2.3 m	Landscape (72-cell)	200